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ubuntu@ip-172-31-10-248:~$ #Lab60. Introduction to SELinux_lab
ubuntu@ip-172-31-10-248:~$ #Objectives: Understand the basics of Security-Enhanced Linux
(SELinux). Learn how to check and manage SELinux status. Change SELinux modes effectively.
Verify SELinux configurations through practical application.
ubuntu@ip-172-31-10-248:~$ sudo apt install policycoreutils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-aws-6.14-headers-6.14.0-1015 linux-aws-6.14-tools-6.14.0-1015 linux-headers-6.14.0-1015-
aws linux-image-6.14.0-1015-aws
  linux-modules-6.14.0-1015-aws linux-tools-6.14.0-1015-aws
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  selinux-utils
The following NEW packages will be installed:
  policycoreutils selinux-utils
0 upgraded, 2 newly installed, 0 to remove and 69 not upgraded.
Need to get 261 kB of archives.
After this operation, 1587 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 selinux-utils
amd64 3.5-2ubuntu2.1 [108 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 policycoreutils
amd64 3.5-2 [153 kB]
Fetched 261 kB in 0s (5718 kB/s)
Selecting previously unselected package selinux-utils.
(Reading database ... 217846 files and directories currently installed.)
Preparing to unpack .../selinux-utils_3.5-2ubuntu2.1_amd64.deb ...
Unpacking selinux-utils (3.5-2ubuntu2.1) ...
Selecting previously unselected package policycoreutils.
Preparing to unpack .../policycoreutils_3.5-2_amd64.deb ...
Unpacking policycoreutils (3.5-2) ...
Setting up selinux-utils (3.5-2ubuntu2.1) ...
Setting up policycoreutils (3.5-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.
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No VM guests are running outdated hypervisor (qemu) binaries on this host.

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ubuntu@ip-172-31-10-248:~$ #Task1:verifySELINUXstatus
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```
ubuntu@ip-172-31-10-248:~$ sestatus
```

```
SELinux status:                disabled
```

```
ubuntu@ip-172-31-10-248:~$ #Key Concepts: SELinux: A set of kernel modifications and user-space tools that add access control security policies to the Linux operating system. Enforcing Mode: SELinux policies are enforced. Permissive Mode: SELinux logs actions that would be blocked in enforcing mode but does not block them.
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ubuntu@ip-172-31-10-248:~$ #Task 2: Change SELinux to Permissive/Enforcing
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```
ubuntu@ip-172-31-10-248:~$ sudo vi /etc/selinux/config
```

```
ubuntu@ip-172-31-10-248:~$ sestatus
```

```
SELinux status:                disabled
```

```
ubuntu@ip-172-31-10-248:~$ cat /etc/selinux/config
```

```
# This file controls the state of SELinux on the system.
```

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# SELINUX= can take one of these three values:
```

```
# enforcing - SELinux security policy is enforced.
```

```
# permissive - SELinux prints warnings instead of enforcing.
```

```
# disabled - No SELinux policy is loaded.
```

```
SELINUX=permissive
```

```
# SELINUXTYPE= can take one of these two values:
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# default - equivalent to the old strict and targeted policies
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# mls      - Multi-Level Security (for military and educational use)
```

```
# src      - Custom policy built from source
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```
SELINUXTYPE=default
```

```
# SETLOCALDEFS= Check local definition changes
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SETLOCALDEFS=0
```

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